

Chapter 38

38 An example: The Otto Cycle

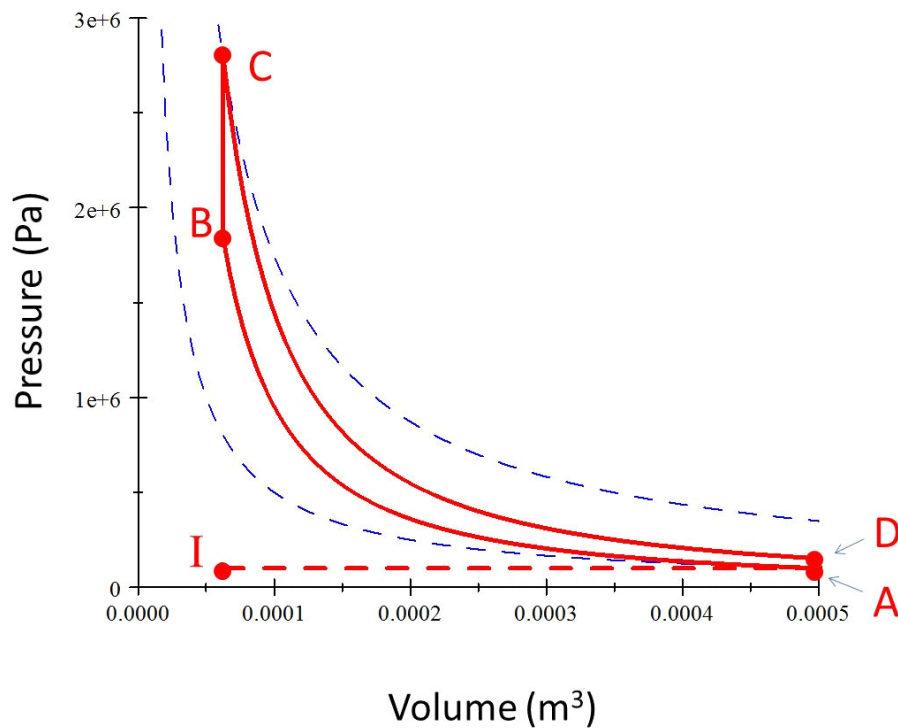
We now know about heat engine. But so far we have been very vague about what happens in the yellow engine circle in our energy transfer diagrams. Let's look at a (mostly) real example of a heat engine. This model of a heat engine is due to a researcher named Otto, so it is called the *Otto Cycle*.

Fundamental Concepts

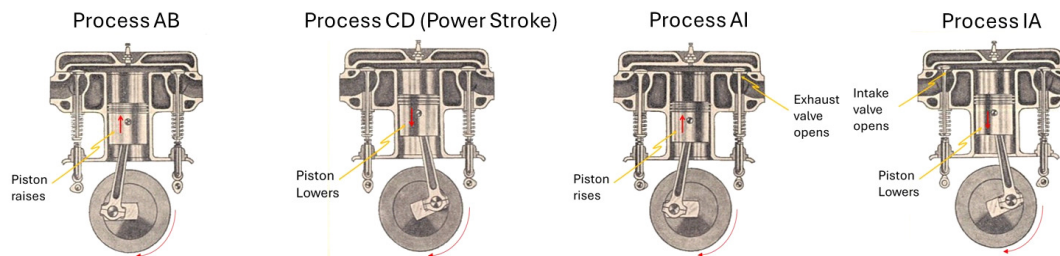
- To describe a heat engine cycle you need to find P , V , n , T , ΔE_{int} , Q , W , for each part of the cycle and W_{eng} , Q_h , Q_c for the entire cycle.
- The Otto Cycle

38.1 An example: The Otto Cycle

A cyclic process is made from more than one thermodynamic process. The Otto cycle technically has six processes. To design an engine we need to describe all of these processes and make sure they work together to produce a net useful work.



The PV diagram above shows the Otto cycle. This cycle is approximately what happens in a gasoline engine, like those in cars or, more like what we find in lawn mowers. It consists of four main processes and two pseudo processes. All of these processes correspond to a piston moving in a cylinder (our favorite thermodynamic system!). Here is a picture of what is happening in the engine.



To understand the Otto cycle let's do the job of finding ΔE_{int} , Q , and w_{int} for each part of the cycle and then describe the state of the gas at the beginning

and end of each cycle by finding P , V , n , T , and E_{int} . For our engine, suppose the compression ratio is

$$\frac{V_A}{V_B} = 8.00$$

Let's take

$$\begin{aligned} V_A &= 500 \text{ cm}^3 = 0.0005 \text{ m}^3 \\ P_A &= 100 \text{ kPa} \\ T_A &= 293 \text{ K} \end{aligned}$$

Then we can find V_B

$$V_B = \frac{V_A}{8} = 62.5 \text{ cm}^3 = 6.25 \times 10^{-5} \text{ m}^3$$

and we will need to measure the temperature at C . Suppose we measure

$$T_C = 1023 \text{ K}$$

and the intake temperature to be

$$T_I = 273 \text{ K}$$

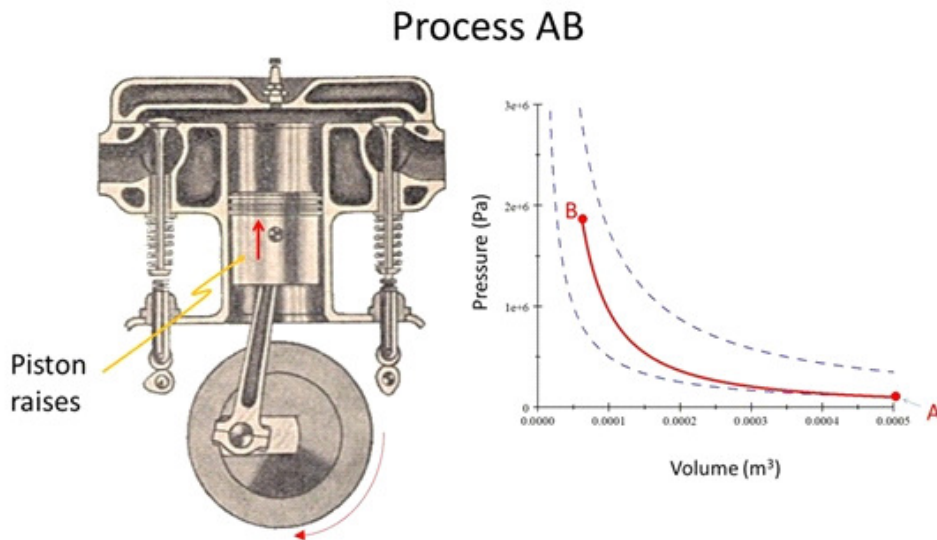
(which probably tells us that this might be a snow blower instead of a lawn mower...but then we are in Rexburg). Further suppose our engine uses a diatomic ideal gas with

$$\begin{aligned} C_V &= \frac{5}{2}R \\ R &= 8.314 \frac{\text{J}}{\text{mol K}} \end{aligned}$$

so

$$\gamma = 1.4$$

Let's follow the process through to see what the state variables are at each point on the PV diagram, and what Q_h and Q_c are for each process, and what the work, W , is and what the change in internal energy will be, ΔE_{int} . Finally, we will find the efficiency of the engine.



Process $A \rightarrow B$ is an adiabatic compression. The piston moves upward, compressing the fuel-air mixture. This happens rapidly, so there is no time for energy to leave by heat, $Q = 0$. That makes it close enough to being adiabatic that we can say this process is approximately adiabatic. In this process $\Delta E_{int} = w_{int}$.

We can use one of our adiabatic equations

$$P_i V_i^\gamma = P_f V_f^\gamma$$

to find the final pressure for this process at B

$$P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$$

But of course this works for any point on the line from A to B so

$$P = P_A \frac{V_A^\gamma}{V^\gamma}$$

We can use this form for P to find the work for process $A \rightarrow B$

$$\begin{aligned}
 w_{AB} &= - \int_{V_A}^{V_B} P dV \\
 &= - \int_{V_A}^{V_B} P_A \frac{V_A^\gamma}{V^\gamma} dV \\
 &= -P_A V_A^\gamma \int_{V_A}^{V_B} \frac{1}{V^\gamma} dV \\
 &= -P_A V_A^\gamma \left(\frac{V^{1-\gamma}}{1-\gamma} \right) \Big|_{V_A}^{V_B} \\
 &= -\frac{P_A V_A^\gamma}{1-\gamma} (V_B^{1-\gamma} - V_A^{1-\gamma})
 \end{aligned}$$

We can rearrange this into a more convenient formula

$$\begin{aligned}
 w_{AB} &= \frac{P_A V_A^\gamma}{\gamma - 1} \left(\frac{1}{V_B^{\gamma-1}} - \frac{1}{V_A^{\gamma-1}} \right) \\
 &= \frac{1}{\gamma - 1} \left(\frac{P_A V_A^\gamma}{V_B^{\gamma-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right) \\
 &= \frac{1}{\gamma - 1} \left(\frac{P_A V_A^\gamma}{V_B^\gamma V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right)
 \end{aligned}$$

And once again using

$$P_B V_B^\gamma = P_A V_A^\gamma$$

or

$$P_A = P_B \frac{V_B^\gamma}{V_A^\gamma}$$

We get

$$w_{AB} = \frac{1}{\gamma - 1} \left(\frac{P_B \frac{V_B^\gamma}{V_A^\gamma} V_A^\gamma}{V_B^\gamma V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right)$$

A lot of things cancel. We get

$$\begin{aligned}
 w_{AB} &= \frac{1}{\gamma - 1} \left(\frac{P_B}{V_B^{-1}} - \frac{P_A V_A^\gamma}{V_A^{\gamma-1}} \right) \\
 &= \frac{1}{\gamma - 1} (P_B V_B - P_A V_A)
 \end{aligned}$$

which we can generalize into yet another basic equation for adiabatic processes!

$$w_{if} = \left(\frac{1}{\gamma - 1} \right) (P_f V_f - P_i V_i)$$

Then for process $A \rightarrow B$ we have

$$w_{AB} = \left(\frac{1}{\gamma - 1} \right) (P_B V_B - P_A V_A)$$

and we can use our equation for P_B that we got before, $P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$ so that our w_{AB} equation becomes

$$w_{AB} = \left(\frac{1}{\gamma - 1} \right) \left(P_A \frac{V_A^\gamma}{V_B^\gamma} V_B - P_A V_A \right)$$

And from our compression ratio equation

$$V_B = \frac{V_A}{8.00}$$

so w_{AB} becomes

$$\begin{aligned} w_{AB} &= \left(\frac{1}{\gamma - 1} \right) \left(P_A \frac{V_A^\gamma}{\left(\frac{V_A}{8} \right)^\gamma} \frac{V_A}{8.00} - P_A V_A \right) \\ &= \left(\frac{1}{\gamma - 1} \right) \left(8^\gamma P_A \left(\frac{V_A}{8} \right) - P_A V_A \right) \\ &= \left(\frac{V_A}{\gamma - 1} \right) (8^{\gamma-1} P_A - P_A) \\ &= \left(\frac{V_A P_A}{\gamma - 1} \right) (8^{\gamma-1} - 1) \end{aligned}$$

So the work is

$$w_{AB} = \left(\frac{V_A P_A}{\gamma - 1} \right) (8^{\gamma-1} - 1)$$

and we know all of these numbers so

$$\begin{aligned} w_{AB} &= \left(\frac{(0.0005 \text{ m}^3)(100 \text{ kPa})}{1.4 - 1} \right) (8^{1.4-1} - 1) \\ &= 162.17 \text{ J} \end{aligned}$$

But this is the work done on the gas, the useful work is the negative of this

$$w_{engAB} = -162.17 \text{ J}$$

Which tells us that it took energy to make the process $A \rightarrow B$ happen. You might need a pull cord to start this engine or have an electric starter motor. The change in internal energy is

$$\Delta E_{AB} = 162.17 \text{ J}$$

and we can find

$$E_{int} = \frac{5}{2}nRT$$

but we need n . We can get it from what we know at A using the ideal gas law

$$\begin{aligned} n &= \frac{P_A V_A}{RT_A} \\ &= \frac{(100 \text{ kPa}) (0.0005 \text{ m}^3)}{\left(8.314 \frac{\text{J}}{\text{mol K}}\right) (293 \text{ K})} \\ &= 2.0525 \times 10^{-2} \text{ mol} \end{aligned}$$

Then

$$\begin{aligned} E_A &= \frac{5}{2}nRT_A \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right) (293 \text{ K}) \\ &= 125.00 \text{ J} \end{aligned}$$

We used

$$P_B = P_A \frac{V_A^\gamma}{V_B^\gamma}$$

but we should find a numeric answer for P_B . Let's use our compression ratio equation for V_B again

$$\begin{aligned} P_B &= P_A \frac{V_A^\gamma}{\left(\frac{V_A}{8}\right)^\gamma} \\ &= P_A \frac{V_A^\gamma}{\frac{V_A^\gamma}{8^\gamma}} \\ &= 8^\gamma P_A \\ &= 8^{1.4} (100 \text{ kPa}) \\ &= 1837.9 \text{ kPa} \end{aligned}$$

And finally, from the ideal gas law

$$\begin{aligned} T_B &= \frac{P_B V_B}{nR} \\ &= \frac{(1837.9 \text{ kPa}) (6.25 \times 10^{-5} \text{ m}^3)}{(2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right)} \\ &= 673.15 \text{ K} \end{aligned}$$

At last we can find E_B

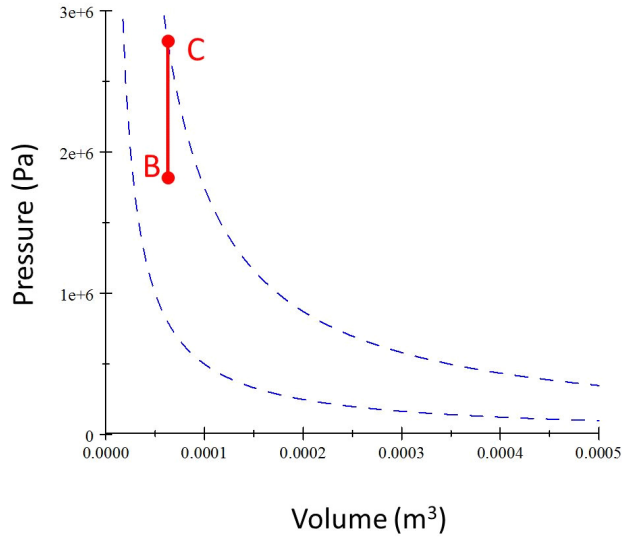
$$\begin{aligned} E_B &= \frac{5}{2}nRT_B \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right) (673.15 \text{ K}) \\ &= 287.17 \text{ J} \end{aligned}$$

what we have learned is summarized in this table

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
A	293	100	0.000 5	2.0525×10^{-2}	125.00	—
B	673.15	1837.9	6.25×10^{-5}	2.0525×10^{-2}	287.17	—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
AB	0	162.17 J	162.17 J	−162.17 J	0	0

That was just one process!

Process $B \rightarrow C$ is an isochoric process.



This is where the spark plug ignites the air-fuel mixture created by the fuel injectors. A great deal of energy is released in a hurry. The temperature jumps up and the pressure does too, but the volume is the same because this process happens in a flash—literally!

So this must be an isochoric process. We expect $w_{BC} = 0$, so $\Delta E_{BC} = Q_{BC}$. The pressure increases, so we expect Q_{BC} to be positive. That means this energy is entering our engine. This must be part of Q_h ! We know the temperature at C and we know the volume $V_C = V_B$ so we can find the final pressure

$$\begin{aligned}
 P_C &= \frac{nRT_C}{V_C} \\
 &= \frac{(2.0525 \times 10^{-2}) (8.314 \frac{\text{J}}{\text{mol K}}) (1023)}{6.25 \times 10^{-5} \text{ m}^3} \\
 &= 2.7931 \times 10^6 \text{ Pa} \\
 &= 2793.1 \text{ kPa}
 \end{aligned}$$

We can find the internal energy at C now

$$\begin{aligned} E_C &= \frac{5}{2}nRT_C \\ &= \frac{5}{2}(2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (1023 \text{ K}) \\ &= 436.42 \text{ J} \end{aligned}$$

We can use

$$\begin{aligned} Q &= nC_V\Delta T \\ &= n\frac{5}{2}R(T_C - T_B) \end{aligned}$$

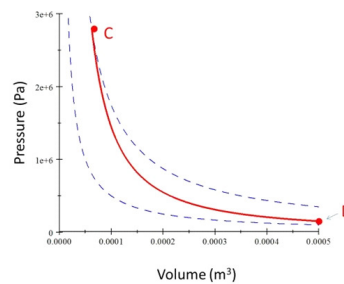
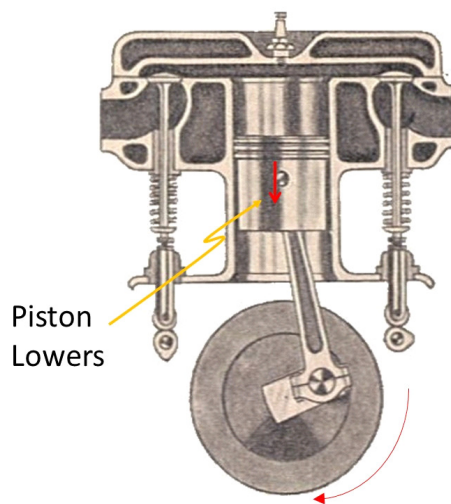
For the process BC to find the energy transferred by heat

$$\begin{aligned} Q_{BC} &= (2.0525 \times 10^{-2} \text{ mol}) \frac{5}{2} \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (1023 \text{ K} - 673.15 \text{ K}) \\ &= 149.25 \text{ J} \end{aligned}$$

We already mentioned that $Q_{BC} = \Delta E_{intBC}$ and $w_{BC} = 0$, so we can complete our table of state variables and process variables for this process.

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
B	673.15	1837.9	6.25×10^{-5}	2.0525×10^{-2}	287.17	—
C	1023	2793.1	6.25×10^{-5}	2.0525×10^{-2}	436.42	—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
BC	149.25	0	149.25	0	149.25	0

Process CD (Power Stroke)



Process $C \rightarrow D$ is another adiabatic process, this time an expansion. This is sometimes called the “power stroke” because the hot, high pressure gas pushes the piston down. This push is what makes the car or lawn mower (snow blower) go. We know that $Q_{CD} = 0$ so $\Delta E_{CD} = w_{CD}$

To find the ideal gas state variables at D let's start with

$$P_i V_i^\gamma = P_f V_f^\gamma$$

again, then

$$P_D = P_C \frac{V_C^\gamma}{V_D^\gamma}$$

but from our PV diagram we see $V_C = V_B$ and $V_D = V_A$. This makes sense because the cylinder and piston system maximum and minimum volumes don't change. So we can write

$$P_D = P_C \frac{V_B^\gamma}{V_A^\gamma} = 8^{-\gamma} P_C$$

and knowing this we can find the temperature at D using the ideal gas law

$$T_D = \frac{P_D V_D}{nR}$$

Let's numerically calculate P_D and T_D at this point.

$$\begin{aligned} P_D &= 8^{-1.4} (2793.1) \\ &= 151.97 \text{ kPa} \end{aligned}$$

and

$$\begin{aligned} T_D &= \frac{(151.97 \text{ kPa}) (0.0005 \text{ m}^3)}{(2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right)} \\ &= 445.28 \text{ K} \end{aligned}$$

And while we are at it we can complete our state D by calculating E_D

$$\begin{aligned} E_D &= \frac{5}{2} n R T_D \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}}\right) (445.28 \text{ K}) \\ &= 189.96 \text{ J} \end{aligned}$$

Now for the work done, we again use our new adiabatic equation

$$\begin{aligned} w_{CD} &= \left(\frac{1}{\gamma - 1}\right) (P_D V_D - P_C V_C) \\ &= \left(\frac{1}{\gamma - 1}\right) (8^{-\gamma} P_C V_A - P_C V_B) \\ &= \left(\frac{P_C}{\gamma - 1}\right) (8^{-\gamma} V_A - V_B) \end{aligned}$$

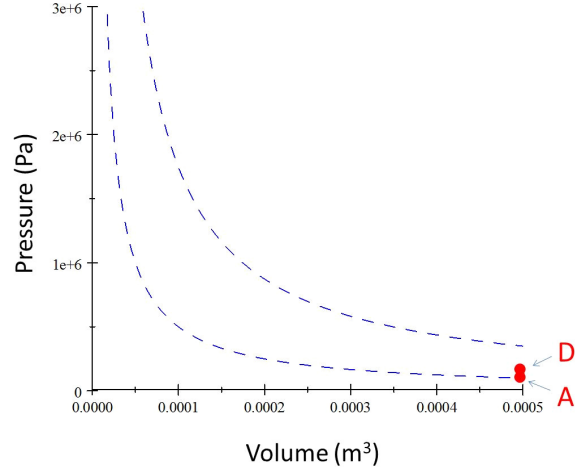
all of which we know, so

$$\begin{aligned} W_{CD} &= \left(\frac{2793.1 \text{ kPa}}{1.4 - 1} \right) (8^{-1.4} (0.0005 \text{ m}^3) - 6.25 \times 10^{-5} \text{ m}^3) \\ &= -246.46 \text{ J} \end{aligned}$$

which completes our set for process CD

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
C	1023	2793.1	6.25×10^{-5}	2.0525×10^{-2}	436.42	—
D	445.28	151.97	0.0005	2.0525×10^{-2}	189.96 J	—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
CD	0	-246.46	-246.46	246.46	0	0

Process $D \rightarrow A$ is an isochoric process. We are nearing the end of the cycle, In process $D \rightarrow A$ a valve opens letting the pressure drop quickly.



Since this happens quickly, the piston does not have time to change position, so the volume does not change. Process DA is (almost) isochoric. We already know P_A , V_A , n , and T_A because that is where we started and this is a cyclic process. But we need to find the process work, heat and change in internal energy. Since the volume does not change, $W_{DA} = 0$ and $\Delta E_{DA} = Q_{DA}$ which we can find using

$$\begin{aligned} Q_{DA} &= nC_V \Delta T \\ &= n \frac{5}{2} R (T_A - T_D) \end{aligned}$$

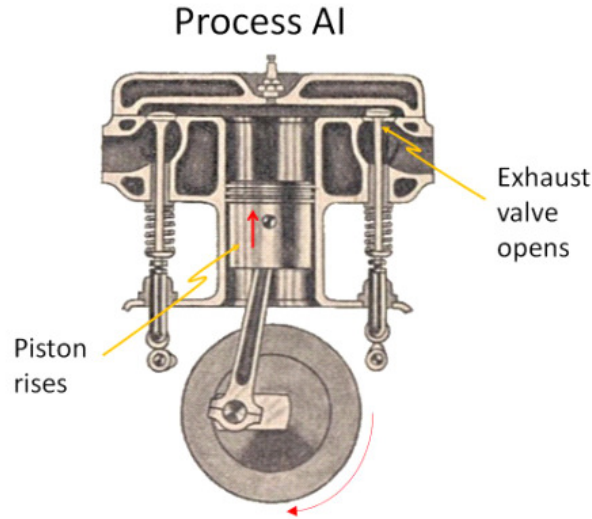
which gives

$$\begin{aligned} Q_{DA} &= (2.0525 \times 10^{-2} \text{ mol}) \frac{5}{2} \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (293 \text{ K} - 445.28 \text{ K}) \\ &= -64.964 \text{ J} \end{aligned}$$

Which tells us that we have lost energy from the engine by heat. So Q_{DA} must contribute to Q_c .

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
D	445.28	151.97	0.000 5	2.0525×10^{-2}	189.96 J	—
A	293	100	0.000 5	2.0525×10^{-2}	125.00	—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
DA	-64.964	0	-64.964	0	0	64.964

Process $A \rightarrow I$ is not a normal thermodynamic process (and neither is process $I \rightarrow A$)



But we are not done. The opening of the valve let the spent fuel begin to leave as exhaust. But we can help remove the exhaust by pushing the piston up. This is done at constant pressure, because the valve is open to the exhaust system air pressure. So this is almost an isobaric process. But it's not quite isobaric because we will lose gas by venting it. The number of moles, n , will change. We can find the new number of moles using the ideal gas law

$$P_I V_I = nRT$$

since $P_I = P_A$ and $V_I = V_B$ because the piston minimum volume is the same each time the piston goes up. Then we can use our compression ratio for V_A and V_I

$$\frac{V_A}{V_I} = 8.00$$

or, solving for V_I

$$V_I = 8/V_A$$

which we could calculate, but it is just the same as V_B

We want n_I , so let's use the ideal gas law for points A and I . We get

$$P_A V_A = n_A R T_A$$

and

$$P_I V_I = n_I R T_I$$

We can divide these two equations

$$\frac{P_A V_A}{P_I V_I} = \frac{n_A R T_A}{n_I R T_I}$$

and note that we have our compression ratio hidden in the left hand side, and recall that $P_I = P_A$ so the left hand side is just 8. And on the right hand side $T_A = T_I$ because we didn't change the temperature of the gas, we just moved it.

$$8 = \frac{n_A}{n_I}$$

We can solve this for the number of moles at piston I

$$n_I = \frac{n_A}{8}$$

And we have all the numbers, so

$$\begin{aligned} n_I &= \frac{(2.0525 \times 10^{-2} \text{ mol})}{8} \\ &= 2.5656 \times 10^{-3} \text{ mol} \end{aligned}$$

And then E_I will be

$$\begin{aligned} E_D &= \frac{5}{2} n_I R T_I \\ &= \frac{5}{2} (2.5656 \times 10^{-3} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (293 \text{ K}) \\ &= 15.625 \text{ J} \end{aligned}$$

The work done for a true isobaric process is

$$w_{AI} = P_A (V_I - V_A)$$

but this assumed that n did not change. Really we are just moving the gas, and it doesn't take much of a force to move the little bit of gas. So to a good approximation we can say

$$\begin{aligned} w_{int} &= F \Delta x \\ &\approx 0 \end{aligned}$$

and we can ignore w_{AI} .

For finding Q_{AI} we would like to use

$$Q = nC_P\Delta T$$

but we quickly run into trouble again because n changed! We are using convection to remove heat instead of conduction. We can get around this by finding ΔE_{int} directly.

$$\begin{aligned}\Delta E_{AI} &= E_I - E_A \\ &= 15.625 \text{ J} - 125.00 \text{ J} \\ &= -109.38 \text{ J}\end{aligned}$$

From this and the first law

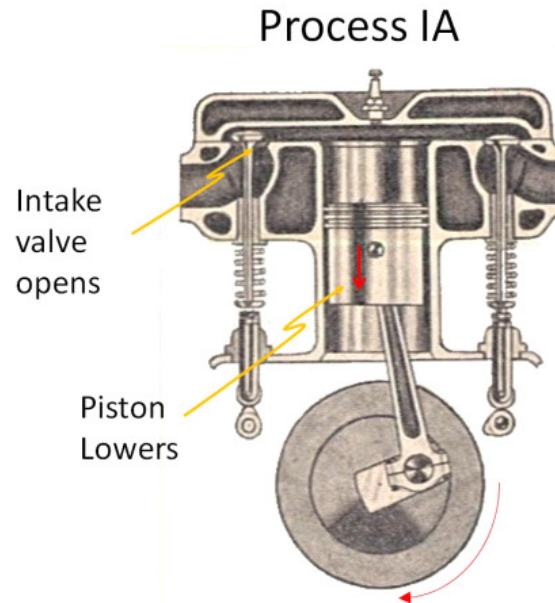
$$\Delta E_{int} = Q + w_{int}$$

We can find Q

$$\begin{aligned}Q &\approx \Delta E_{int} \\ &\approx -109.38 \text{ J}\end{aligned}$$

Energy is leaving with the gas, so we seem to pick up a contribution to Q_c . But really this is not part of our thermodynamic process, because we mechanically removed the energy by removing the gas. Let's denote this by putting our energy in parenthesis to remind us that this is a more mechanical than thermal process.

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
A	293	100	0.000 5	$2.052\,5 \times 10^{-2}$	125.00	—
I	293	100	6.25×10^{-5}	$2.565\,6 \times 10^{-3}$	15.625 J	—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
AI	(−109.38 J)	~ 0	(−109.38)	~ 0	0	(109.38)

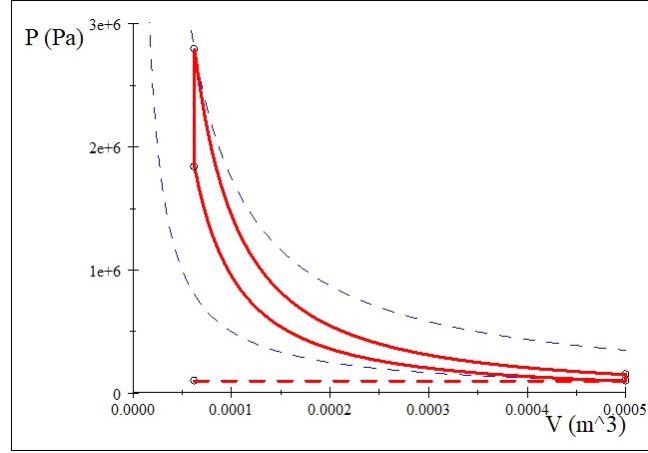


Now the exhaust valve closes, and the intake valve opens. The new fuel mixture is brought in. We essentially reverse process AI in process IA .

State	T (K)	P (kPa)	V (m^3)	n (mol)	E_{int} (J)	—
I	293	100	6.25×10^{-5}	2.5657×10^{-3}	15.625 J	—
A	293	100	0.0005	2.0525×10^{-2}	125.00	—
Process	Q (J)	w_{int} (J)	ΔE_{int} (J)	w_{eng} (J)	Q_h	Q_c
IA	(109.38 J)	~ 0	(109.38 J)	~ 0	(−109.38 J)	0

But there is a difference. The new gas has useful fuel, and that means we are ready to complete the cycle again by compressing the gas, igniting it, and receiving the power stroke and exhaust. Note that Processes AI and IA have a net zero effect thermodynamically, but are very important to making your car or lawn mower go!

Putting it all together gives our PV diagram for the whole cycle. Note that since n changed for the $AI - IA$ processes, it is not really possible to include it on the PV diagram. That is why it has a dashed line. We know the volume and pressure along the path, but the predicted temperature from the graph may not be right because n changed too.



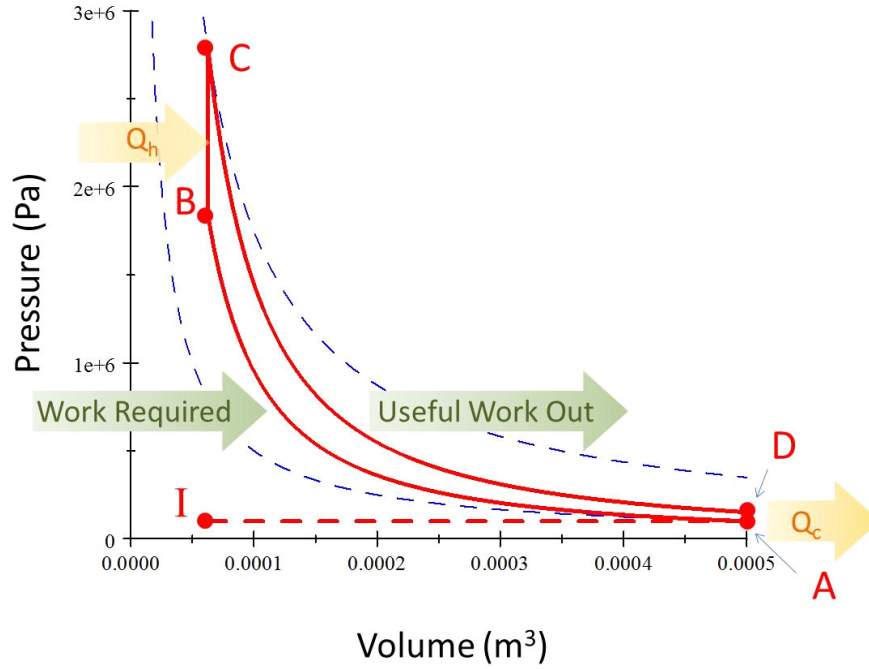
Let's summarize what we found in large tables. First the ideal gas state variables and E_{int}

State	T (K)	P (kPa)	V (m ³)	n (mol)	E_{int} (J)
A	293	100	0.000 5	$2.052\,5 \times 10^{-2}$	125.00
B	673.15	1837.9	6.25×10^{-5}	$2.052\,5 \times 10^{-2}$	287.17
C	1023	2793.1	6.25×10^{-5}	$2.052\,5 \times 10^{-2}$	436.42
D	445.28	151.97	0.000 5	$2.052\,5 \times 10^{-2}$	189.96 J
I	293	100	6.25×10^{-5}	$2.565\,6 \times 10^{-3}$	15.625

and now the values of Q , w , and E_{int} for the processes. We have included w_{eng} , Q_h and Q_c as well

Process	Q (J)	w_{int} (J)	ΔE_{int} (J)	w_{eng} (J)	Q_h	Q_c
AB	0	162.17	162.17	-162.17	0	0
BC	149.25	0	149.25	0	149.25	0
CD	0	-246.46	-246.46	246.46	0	0
DA	-64.964	0	-64.964	0	0	64.964
AI	(-109.38)	(~ 0)	(-109.38)	(~ 0)	(0)	(109.38)
IA	(109.38)	(~ 0)	(109.38)	(~ 0)	(-109.38)	(0)
Whole Cycle	84.286	-84.29	0	84.29	149.25	64.964

Note that the last row of the table is a total for the entire cycle. We see that we have useful work of 84.3 J and that $Q_h = 258.63$ J and $Q_c = 174.34$. We can use this to identify the processes that make Q_h and Q_c happen on our graph. The useful work is the difference between the work done in the two adiabatic processes.



We can see that it takes some work to make process AB happen. And recall that that is why we need a starter motor or a pull cord to get our engine started. Once the engine gets going the momentum from the piston and crank shaft keep the engine running. But still, process AB requires energy input by work, so it is like a resistance for the cycle.

What is the efficiency of this cycle? We don't have a really good way with our simple view of thermodynamics to accurately account for the change in chemical potential energy involved with state I . The energy transfer in processes AI and IA are, really mechanical energy processes. So for this class it might be more fair to say that processes AI and IA have no net effect (except for refueling) and exclude their effect from Q_c and Q_h then ignoring the I processes

$$\begin{aligned}\eta &= \frac{w_{eng}}{Q_h} \\ &= \frac{84.29}{149.25} \\ &= 0.56476\end{aligned}$$

For a gasoline engine, it is often quoted that the efficiency is

$$\eta_{otto} = 1 - \frac{1}{\left(\frac{V_A}{V_B}\right)^{\gamma-1}}$$

Let's test this.

$$\begin{aligned}\eta_{otto} &= 1 - \frac{1}{\left(\frac{0.0005}{6.25 \times 10^{-5}}\right)^{1.4-1}} \\ &= 0.56472\end{aligned}$$

which is nearly the same. But you might guess that this is very optimistic. Efficiencies for actual engines come in at values around 20%.

This was a lot of work, but then engine design is a lot of work! And that is what they pay engineers to do.

But you might wonder, if an engine can't be 100% efficient, just how efficient can an engine be? If you were a mechanical engineer (or these days an electrical engineer making electric vehicles), it would be good to know if your design was unreasonable before you start manufacturing! Let's take that on in our next lecture.